

Geophysical Research Letters®



RESEARCH LETTER

10.1029/2026GL121809

Key Points:

- High resolution seismic imaging reveals a clear east–west contrast in upper mantle structure
- Low velocity anomalies in the east are partly explained by the influence of mantle anisotropy
- Vertically elongated high velocity structures in the west suggest ongoing lithospheric modification

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

Y. Kim,
youngkim@snu.ac.kr

Citation:

Lee, H., Kim, Y., Bezada, M. J., & Lee, H. (2026). Seismic imaging reveals ongoing modification of craton margins in Northeast Asia. *Geophysical Research Letters*, 53, e2026GL121809. <https://doi.org/10.1029/2026GL121809>

Received 15 JAN 2026

Accepted 23 APR 2026

Author Contributions:

Conceptualization: Hwaju Lee, YoungHee Kim, Maximiliano J. Bezada
Data curation: Hwaju Lee, YoungHee Kim
Formal analysis: Hwaju Lee, YoungHee Kim
Funding acquisition: YoungHee Kim
Investigation: Hwaju Lee, YoungHee Kim, Hyunwoo Lee
Methodology: Hwaju Lee, YoungHee Kim, Maximiliano J. Bezada
Project administration: YoungHee Kim
Resources: Hwaju Lee, YoungHee Kim
Software: Hwaju Lee, Maximiliano J. Bezada
Supervision: YoungHee Kim
Validation: Hwaju Lee, YoungHee Kim, Maximiliano J. Bezada, Hyunwoo Lee

© 2026. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs License](#), which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Seismic Imaging Reveals Ongoing Modification of Craton Margins in Northeast Asia

Hwaju Lee¹ , YoungHee Kim^{1,2} , Maximiliano J. Bezada³ , and Hyunwoo Lee^{1,2} 

¹Center for Deep-Surface Coupling of Earth, Seoul National University, Seoul, Republic of Korea, ²School of Earth and Environmental Sciences, Seoul National University, Seoul, Republic of Korea, ³Department of Earth & Environmental Sciences, University of Minnesota, Minneapolis, MN, USA

Abstract Cratonic margins are commonly modified, yet the processes responsible remain debated. The Korean Peninsula, situated between adjacent cratons, the Japan Trench, and a back-arc system, provides a natural setting to investigate this problem. Using 4 years of dense seismic observations, we image upper-mantle structure beneath the peninsula with P-wave tomography that incorporates a prescribed anisotropy model to reduce velocity artifacts. The results reveal a pronounced east–west contrast, with slow velocities beneath the eastern peninsula and fast velocities beneath the western peninsula. The eastern low-velocity anomalies are consistent with hot and/or compositionally modified mantle, whereas the fast western anomalies may indicate ongoing lithospheric modification beneath the peninsula.

Plain Language Summary Cratons are the oldest and most stable parts of continents, but their edges can be modified by deep Earth processes. Using dense seismic observations across the Korean Peninsula, we created a high-resolution image of the upper mantle that accounts for the directional behavior of seismic waves. Our results show a strong contrast between the eastern and western peninsula, with slower seismic speeds in the east and faster speeds in the west. The eastern anomalies are partly explained by seismic anisotropy, and their remaining cause may involve temperature or water, but cannot be uniquely determined. In contrast, the western anomalies indicate that the lithosphere is being modified at depth, even though surface features remain largely unchanged, suggesting that the craton margin beneath the Korean Peninsula is actively modified.

1. Introduction

Cratons represent the most stable portions of the continental lithosphere, yet their margins are often sites of profound and ongoing modification. Globally, diverse observations highlight craton instability: in Africa, plume activity and rifting have thinned the Tanzanian craton (Celli et al., 2020); in North America, the Wyoming and Colorado Plateau margins show evidence for delamination of cratonic lithosphere (e.g., Levander et al., 2011; Sine et al., 2008), while seismic studies beneath the Appalachians indicate mantle upwelling along the craton margin (e.g., Byrnes et al., 2019); and in the Canadian Cordillera, inherited sutures appear to guide lithospheric removal (e.g., Chen et al., 2019). Despite these observations, the processes that destabilize craton margins—particularly where subduction, extension, and mantle flow interact—remain debated.

The Korean Peninsula (KP) offers a natural laboratory to address this problem. The KP, with its Archean outcrops (e.g., Chough et al., 2000), sits at the junction of the North China Craton (NCC), the South China Craton (SCC), and the Pacific subduction system, where cratonic lithosphere interacts with back-arc extension and subduction-driven flow. The East Sea, a back-arc basin formed by spreading between Korea and Japan (e.g., Tatsumi et al., 1989; Van Horne et al., 2017), separates the KP from the Japan Trench and records extension along the KP's eastern margin (Figure 1a).

Meanwhile, the NCC has long served as a canonical example in studies for investigating craton destruction mechanisms (e.g., He et al., 2015; Wu et al., 2019; Zhu et al., 2012). Proposed explanations for its lithospheric thinning vary among studies but are often related to 1) thermal erosion by mantle plume (Polat et al., 2006; Zhou & Zhai, 2022) or intraplate volcanism associated with the big mantle wedge (BMW) system (He, 2015; Tian & Zhao, 2013; Zhu & Xu, 2019), and/or 2) mechanical erosion by delamination due to extension (Wang et al., 2022; Zhang et al., 2014) and/or flat subduction and its retreat of the Paleo-Pacific Plate (Kusky et al., 2014; Lei, 2012; Wu et al., 2019). By contrast, the SCC, which is adjacent to the NCC, has experienced lithospheric

Visualization: Hwaju Lee,
YoungHee Kim

Writing – original draft: Hwaju Lee,
YoungHee Kim

Writing – review & editing: Maximiliano
J. Bezada, Hyunwoo Lee

thinning especially in its eastern part (i.e., Yangtze Block), yet lacks the widespread or severe destruction as seen in the NCC (e.g., Li et al., 2015; Zhang et al., 2023).

Although the destruction mechanisms remain debated, multiple observations indicate eastward lithospheric thinning toward the KP. Receiver function studies show progressive shoaling of both the Moho and lithosphere–asthenosphere boundary (LAB), reaching depths of ~30 km and 60–70 km in the easternmost NCC and SCC (e.g., Chen, 2010; Dong et al., 2022; He et al., 2015; Zhang et al., 2019, 2023). Gravity, geoid, and topographic constraints support this trend, and helium isotopes suggest a shallow LAB beneath the East Sea (e.g., Lee et al., 2025; Xia et al., 2020; Xu et al., 2016).

The KP is among the best-instrumented continental margins, with a dense broadband network enabling high-resolution imaging of lithospheric structure (Figures S1a–S1b in Supporting Information S1). We exploit this coverage to invert for teleseismic P-wave tomography that explicitly incorporates mantle seismic anisotropy, thereby reducing anisotropy-related velocity artifacts and improving the geologic interpretability of the resulting model. Our images resolve upper-mantle structure beneath the KP and provide new insight into how stable lithosphere can be modified along craton margins.

2. Data and Methods

2.1. Isotropic Tomography Inversion

We analyzed 4 years (2018–2021) of teleseismic P-wave data ($M_w \geq 5.5$) recorded at 140 stations across South Korea, including offshore islands (Figures S1a–S1b in Supporting Information S1). The average spacing between inland stations is ~30 km. A total of 43,977 travel times from 399 events at epicentral distances of 30–90° were inverted. P-wave arrivals were manually picked using three band-pass filters (0.2, 0.4, and 1.0 Hz), and multi-channel cross-correlation (VanDecar & Crosson, 1990) was applied to compute relative travel-time delays for isotropic tomography. The AK135 model (Kennett et al., 1995) was used as the reference velocity model. We employed a hybrid ray-tracing approach that combines conventional ray theory with finite-frequency sensitivity kernels (Bezada et al., 2013). Damping and smoothing parameters were determined through L-curve analysis. Since the inversion uses a large number of observations, random picking errors are expected to be reduced through the regularization. Model resolution was evaluated using checkerboard, hit-quality, and synthetic tests (Figures S2–S4 in Supporting Information S1), indicating that lateral anomalies of $\sim 50 \times 50 \text{ km}^2$ and vertical scales of ~70 km are well resolved to depths of ~300 km. Previous studies suggest that lateral variations in crustal velocity structure may produce travel-time delays of up to ~0.5 s (e.g., Chang & Baag, 2005), which are relatively small compared to the overall travel-time residuals. Nevertheless, we apply station and event corrections to further mitigate the effects of near-surface structure and event-dependent travel-time variations. These parameters were held fixed during the inversion.

2.2. Tomographic Inversion With Prescribed Anisotropy

Although recent studies have increasingly adopted anisotropic body-wave tomography (VanderBeek & Faccenda, 2021), isotropic inversion remains the standard approach, as introducing anisotropy (both its direction and magnitude) into the inversion makes it even less stable. Moreover, neglecting anisotropy can introduce artificial velocity anomalies, potentially leading to erroneous interpretation of the velocity structure (e.g., Bezada et al., 2016; Lee et al., 2023; VanderBeek & Faccenda, 2021). In this study, we used a method to incorporate anisotropy by introducing hypothetical anisotropy (i.e., a prescribed anisotropy model) as an a priori constraint in the inversion (Bezada et al., 2016). Travel-time delays are corrected by removing the delays introduced by the prescribed anisotropy during the forward calculation, and the anisotropy field is held fixed throughout all inversion iterations (Lee et al., 2023).

2.3. Prescribed Anisotropy Models for the Korean Peninsula

To investigate mantle dynamics beneath the KP while minimizing potential artificial velocity anomalies caused by mantle anisotropy, we developed hypothetical anisotropy models based on previous study (Song et al., 2020), shear-wave splitting (SWS) measurements (Kang & Shin, 2009), and the volume of low velocity anomalies (LVAs) and high velocity anomalies (HVAs) in the isotropic tomography result for the eastern and western KP, respectively (Figures 1b–1j, Figures S5 and S6 in Supporting Information S1). Previous studies and our isotropic

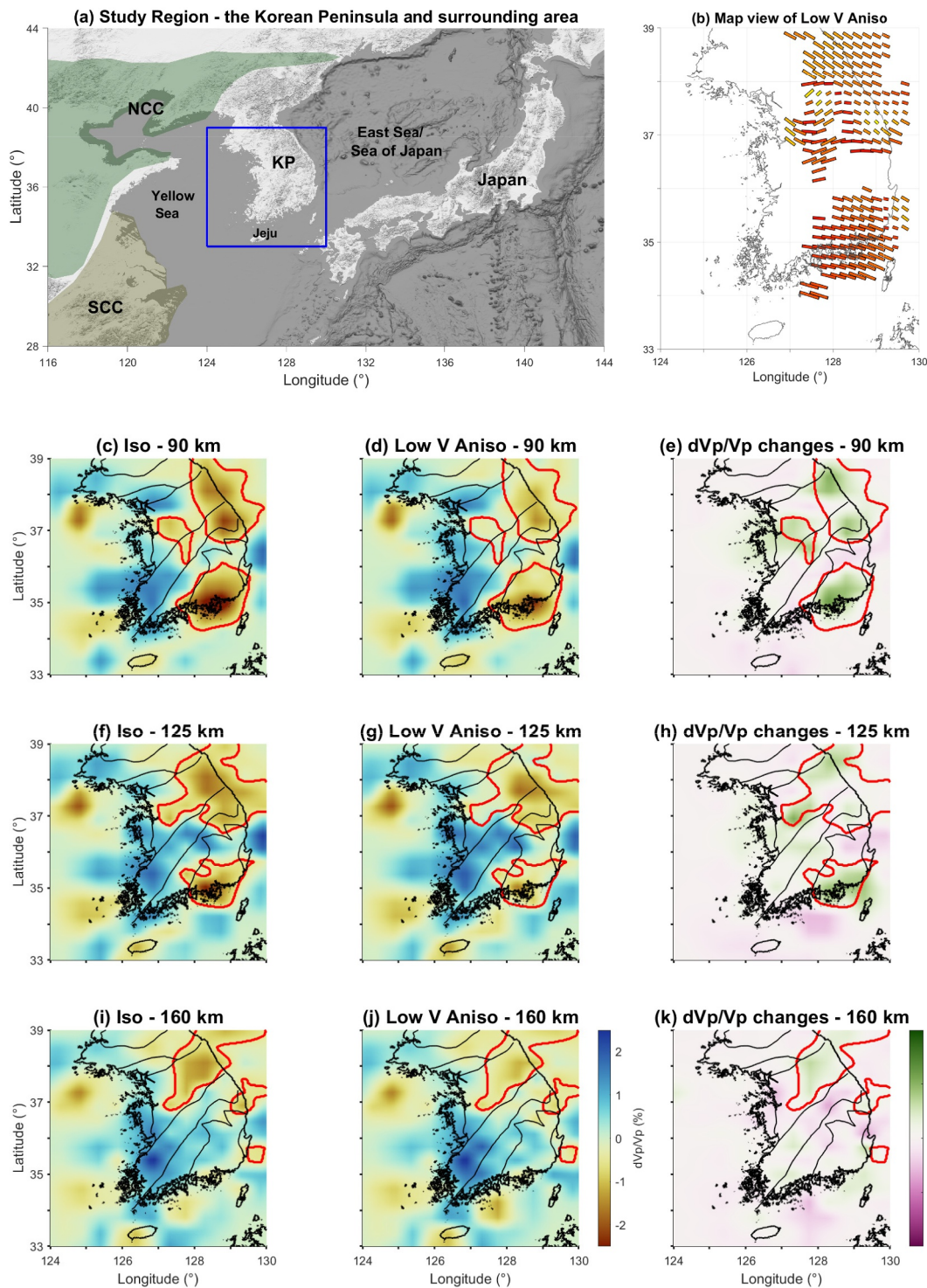


Figure 1. (a) Geological setting of the study region showing the Korean Peninsula (KP) and surrounding areas (blue box). White areas indicate land above sea level, while gray areas indicate land below sea level. The green and brown shaded regions indicate the extent of the North China Craton (NCC) and South China Craton (SCC), respectively. (b) Map view of the anisotropy model (Low V Aniso) incorporated in panels (d, g, j). The length of each bar represents the strength of anisotropy, and its orientation indicates the direction of anisotropy, prescribed to reproduce the shear-wave splitting observations associated with the low-velocity anomalies. Further details for both anisotropy models (Low V Aniso and High V Aniso) are provided in Figure S7 in Supporting Information S1. (c–k) Comparison of isotropic and anisotropic tomographic models highlighting low-velocity anomalies (low velocity anomalies) beneath the eastern KP. Red contours outline the area where the prescribed low-velocity anisotropy model (Low V Aniso) is applied. Isotropic inversion results at depths of 90, 125, and 160 km are shown in (c, f, i), and corresponding anisotropic inversion results in (d, g, j). Panels (e, h, k) show differences in dV_p/V_p , where positive values indicate reduced low-velocity amplitudes in the anisotropic models relative to the isotropic results. Black solid lines within the peninsula delineate major geological boundaries within the Korean Peninsula (KP).

tomography consistently reveal LVAs beneath the topographically elevated eastern KP (Park & Hong, 2024; Song et al., 2020) and HVAs beneath the relatively flat western KP (Song et al., 2020).

The LVAs in the eastern and HVAs in the western Korean Peninsula likely reflect a combination of factors, including mantle temperature, hydration state, and volatile content. Nevertheless, we adopt a simplified end-member assumption that these anomalies arise solely from mineralogical anisotropy to evaluate the potential contribution of anisotropy to the observed velocity structure. We assume that mantle anisotropy arises exclusively from the A-type olivine fabric that aligns the a-axis with the direction of mantle flow (Ismail & Mainprice, 1998). For instance, horizontally-oriented seismically fast directions (SFD) may generate artificial LVAs in the velocity model, whereas vertically aligned fast directions may produce apparent HVAs (e.g., Bezada et al., 2016; VanderBeek & Faccenda, 2021). If the prescribed anisotropy model approximates the true mantle anisotropy, the inversion should suppress the artifacts and reduce the model norm (Bezada et al., 2016; Lee et al., 2021, 2023; VanderBeek & Faccenda, 2021).

We constructed two independent anisotropy models targeting the eastern and western KP to evaluate the contribution of anisotropy in the velocity anomalies (Figure 1b and Figure S7 in Supporting Information S1). For the eastern LVAs, we defined the volume of LVAs $< -0.5\%$ P-wave velocity perturbation (dV_p/V_p) as the volume of prescribed anisotropy model (red solid contour in Figures 1c–1k, Figures S7a and S7c in Supporting Information S1). The field (i.e., magnitude) and orientation of the anisotropy model are set to reproduce delay times and fast polarization directions observed in the SWS study, respectively (Kang & Shin, 2009). For the western HVAs, we defined the volume of HVAs $> +0.5\%$ dV_p/V_p as the volume of prescribed anisotropy model (Figures S7b and S7d in Supporting Information S1, and blue solid contour in Figure S8 in Supporting Information S1). A previous study reported small SWS times with average less than 0.5 s in western KP (Kang & Shin, 2009). However, vertically aligned SFD does not produce SWS, we assume that the observed HVAs $> +0.5\%$ dV_p/V_p in the isotropic tomography for the western KP solely come from the anisotropy whose SFD are vertically aligned. The anisotropy models for the eastern and western Korean Peninsula were not applied simultaneously in a single forward calculation. Instead, each model was implemented and tested independently to evaluate its effect on the inversion results. As a result, the delay corrections are path-dependent and vary by station, rather than being uniform.

3. Results

3.1. Isotropic Tomography

Our isotropic P-wave tomography model reveals clear east–west variations in upper mantle velocity structure beneath the KP, characterized by LVAs beneath the mountainous eastern region and HVAs beneath the relatively flat western region (Figures 1c–1f and 1i, Figures S5 and S6 in Supporting Information S1). These results are broadly consistent with the previous velocity model (Song et al., 2020) (Figure S9 in Supporting Information S1). We find that the pronounced east–west velocity contrast does not align with the major boundaries based on geological surface expressions (Kim, 2002), which are generally oriented northeast–southwest. This mismatch may indicate that present-day crustal deformation primarily reflects inherited tectonic structures rather than present mantle-driven modification.

3.1.1. LVAs in Eastern KP

The LVAs in eastern KP can be divided into two groups: the northeast anomalies, which extend from the surface down to ~ 300 km depth, and the southeast anomalies, which reach depths of ~ 130 km. Compared to the previous model (Song et al., 2020), our tomographic model reveals stronger southeast LVAs (Figure S9 in Supporting Information S1). In particular, the southeast region of the KP exhibits up to -2.5% dV_p/V_p in our model, whereas the prior model showed anomalies smaller than -1% in this region.

3.1.2. HVAs in Western KP

The HVAs in western KP extend to a depth of ~ 200 km. Particularly, the HVAs in the southernmost part of western KP (i.e., around 126° – 127° E, 34.5° – 35.5° N) extend to depths exceeding 250 km (Figures S5–S6 in Supporting Information S1).

3.2. Prescribed Anisotropy Tomography

When the prescribed anisotropy models were incorporated independently into the tomographic inversion, both models provided a good fit to the travel-time data. However, their normalized model norms, relative to the isotropic model, differ.

3.2.1. LVAs in Eastern KP

Incorporating the prescribed anisotropy model into the isotropic inversion leads to a reduction in both the magnitude and spatial extent of the LVAs. The red contour lines in Figures 1c–1k delineate the regions where prescribed anisotropy model is placed. These lines correspond to an area with isotropic dV_p/V_p of $< -0.5\%$ in each depth slice, serving as a reference for comparison.

In the isotropic tomography results, LVAs show velocity perturbations of up to $\sim 2.5\%$ slower, whereas the prescribed anisotropy tomography generally reveals LVAs with perturbations less than 1.5% . The reduction is particularly prominent in the southern LVAs, where we find up to 2% reduction in the LVAs (Figures 1c–1k and Figure S10 in Supporting Information S1). Consequently, the normalized model norm relative to the isotropic model is reduced by approximately 11% across the entire KP, indicating that a significant portion of the original LVAs may be attributed to neglected anisotropy (Figures S10–S12a, S12b, and S12d in Supporting Information S1).

3.2.2. HVAs in Western KP

In contrast, introducing the prescribed anisotropy model for the HVAs in western KP results in only minor changes. The blue contour lines in Figure S8 in Supporting Information S1 denote the HVA boundaries that are $> +0.5\%$ dV_p/V_p from the isotropic tomography at each depth slice. Accordingly, insignificant changes in the model norm are found (Figures S8, S10, S12a, S12c, S12e, and S13 in Supporting Information S1). Yet, we find a minor reduction in the velocity anomalies in the HVAs (Figure S14 in Supporting Information S1). For instance, we do find 0.5% reduction patches in the HVAs at depths of 125 and 160 km, particularly near the southern edge of the western KP, where HVAs are found (Figure S14 in Supporting Information S1).

4. Discussion

4.1. Eastern Korean Peninsula: LVAs and Anisotropy

4.1.1. Influence of Anisotropy on Imaged LVAs

The reduction of LVAs observed in prescribed anisotropy tomography provides evidence that existing anisotropy is affecting the imaged amplitudes of velocity anomalies in the isotropic tomography results. The reductions in both the LVAs (up to $\sim 2\%$) and the model norm ($\sim 11\%$) suggest that our anisotropy model closely approximates the actual anisotropic structure of the mantle in this region (Figures 1c–1k, Figures S10–S12b, and S12d in Supporting Information S1). These results imply that part of the LVAs imaged in the isotropic tomography may be anisotropic artifacts, leading to an overestimation of their amplitude and volume.

However, including the anisotropy model in the inversion does not fully remove the LVAs, indicating that residual anomalies likely reflect intrinsic mantle properties, most plausibly elevated temperature and/or hydration. Similar associations between low-velocity anomalies and thermal upwelling or mantle hydration have been reported beneath East Africa and intraplate volcanoes in Northeast Asia (e.g., Celli et al., 2020; Yang & Facenda, 2020). Meanwhile, seismic evidence for lithospheric thinning and elevated heat in the northern Canadian Cordillera (e.g., Chen et al., 2019) and the Northern Appalachians (e.g., Byrnes et al., 2019) provides additional analogs. Below, we evaluate end-member scenarios in which LVAs are attributed to a single physical parameter and assess how inferred temperature or hydration changes when SWS-constrained anisotropy is included.

4.1.2. Origin of LVAs

LVAs in seismic tomography are commonly linked to elevated mantle temperatures due to the strong thermal sensitivity of seismic velocities (e.g., Cammarano et al., 2003; Karato, 1993). Previous studies have suggested mantle upwelling as a source of the LVAs in the eastern KP (Park & Hong, 2024; Song et al., 2020). Our tomographic model also shows a strong spatial correlation with both the distribution of Cenozoic volcanoes in the

KP and locations where helium isotope ratios suggest a mantle source of helium (Figure S15 in Supporting Information S1) (Kim et al., 2024).

To quantify mantle temperature and evaluate the potential for partial melt, we converted LVA amplitudes to temperature estimates at 90 and 125 km depth (Figure S16 in Supporting Information S1), assuming it is purely coming from thermal origin. Based on the relationship between V_p and temperature (T)— $\partial \ln V_p / \partial T$ (Cammarano et al., 2003), we assumed pressures of 3 and 4 GPa and ambient mantle temperatures of 1250°C and 1370°C at depths of 90 and 125 km, respectively, representing typical continental margin conditions (Lee et al., 2011). We used 1400°C and 1500°C as the dry solidus temperatures of peridotite at 3 and 4 GPa, respectively (Hirschmann, 2000). Further details on the estimation can be found in the Supplementary Information (Figure S15 in Supporting Information S1).

When comparing temperature estimates from isotropic and prescribed anisotropy models at 90 and 125 km depths, we find a lower potential for partial melt in the prescribed anisotropy model given the reduction of LVAs. At 90 km depth, the isotropic model suggests possible partial melting in both the northeast and southeast of the KP, whereas the prescribed anisotropy model indicates the presence of melt is likely restricted to the southeast region (Figure S16 in Supporting Information S1). At 125 km depth, the potential melt regions are similar between the two models; however, the volume of inferred melt is smaller in the prescribed anisotropy model result. These differences highlight how neglecting anisotropy can overestimate thermal anomalies and melt extent.

4.2. Western Korean Peninsula: HVAs and LAB of Adjacent Cratons

4.2.1. Limited Influence of Anisotropy on HVAs

Unlike the LVAs in the east, the HVAs beneath the western KP remain largely unchanged when anisotropy is incorporated. The limited changes in both the HVAs and the whole-model norm in the prescribed anisotropy tomography suggest that these anomalies are unlikely to arise from anisotropy with vertically aligned SFDs. When the assumed anisotropy model deviates strongly from the true mantle structure, it commonly introduces artifacts and increases the model norm (Bezada et al., 2016; Lee et al., 2021, 2023). However, we observe no such increase in either velocity anomalies or model norms (Figures S8, S10, S12c, S12e, and S13 in Supporting Information S1). The minor norm reductions and slight weakening of HVAs at depths of 125–160 km (Figure S14 in Supporting Information S1) may indicate localized vertically oriented anisotropy, although its influence appears limited.

4.2.2. LAB of Adjacent Cratons

Although the cratonic lithosphere extends toward the KP (e.g., Chough et al., 2000), interpreting the HVAs beneath the western KP as intact cratonic lithosphere at depths >160 km is difficult to reconcile with regional LAB constraints. Receiver function and gravity studies indicate that the LAB beneath the NCC and SCC thins eastward toward the Korean Peninsula, reaching depths of ~60 km in eastern China (e.g., Chen, 2010; Dong et al., 2022; He et al., 2015; Xia et al., 2020; Xu et al., 2016; Zhang et al., 2019, 2023). Tomographic studies further show that the lithospheric base beneath the eastern NCC is ~70 km deep, significantly shallower than in the central and western NCC, where it extends to ~150 km (e.g., Feng et al., 2022; Tian & Zhao, 2011; Xiao et al., 2023). At greater depths, high-velocity bodies imaged beneath eastern China are commonly interpreted as delaminated lithospheric fragments rather than coherent cratonic roots (Dong et al., 2022; Feng et al., 2022; Lei, 2012; Tian & Zhao, 2011). Similar lithospheric thinning and alteration have also been suggested beneath the eastern SCC (Xiao et al., 2023; H. Zhang et al., 2023). Such craton modification is not unique to Northeast Asia (e.g., He & Kapp, 2023; Levander et al., 2011; Sine et al., 2008; West et al., 2009; Zandt et al., 2004).

4.3. Tectonic Implications for the Western Korean Peninsula

4.3.1. Evaluation of Ongoing Lithospheric Modifications

Our high-resolution prescribed-anisotropy tomography may suggest ongoing lithospheric modification beneath the western KP. Vertically elongated HVAs are widely interpreted as signatures of hanging lithosphere and/or delamination. Similar structures documented beneath the Colorado Plateau, Great Basin, and southern Sierra Nevada in the western United States have been attributed to delaminated and downwelling lithosphere (He & Kapp, 2023; Levander et al., 2011; Sine et al., 2008; West et al., 2009; Zandt et al., 2004).

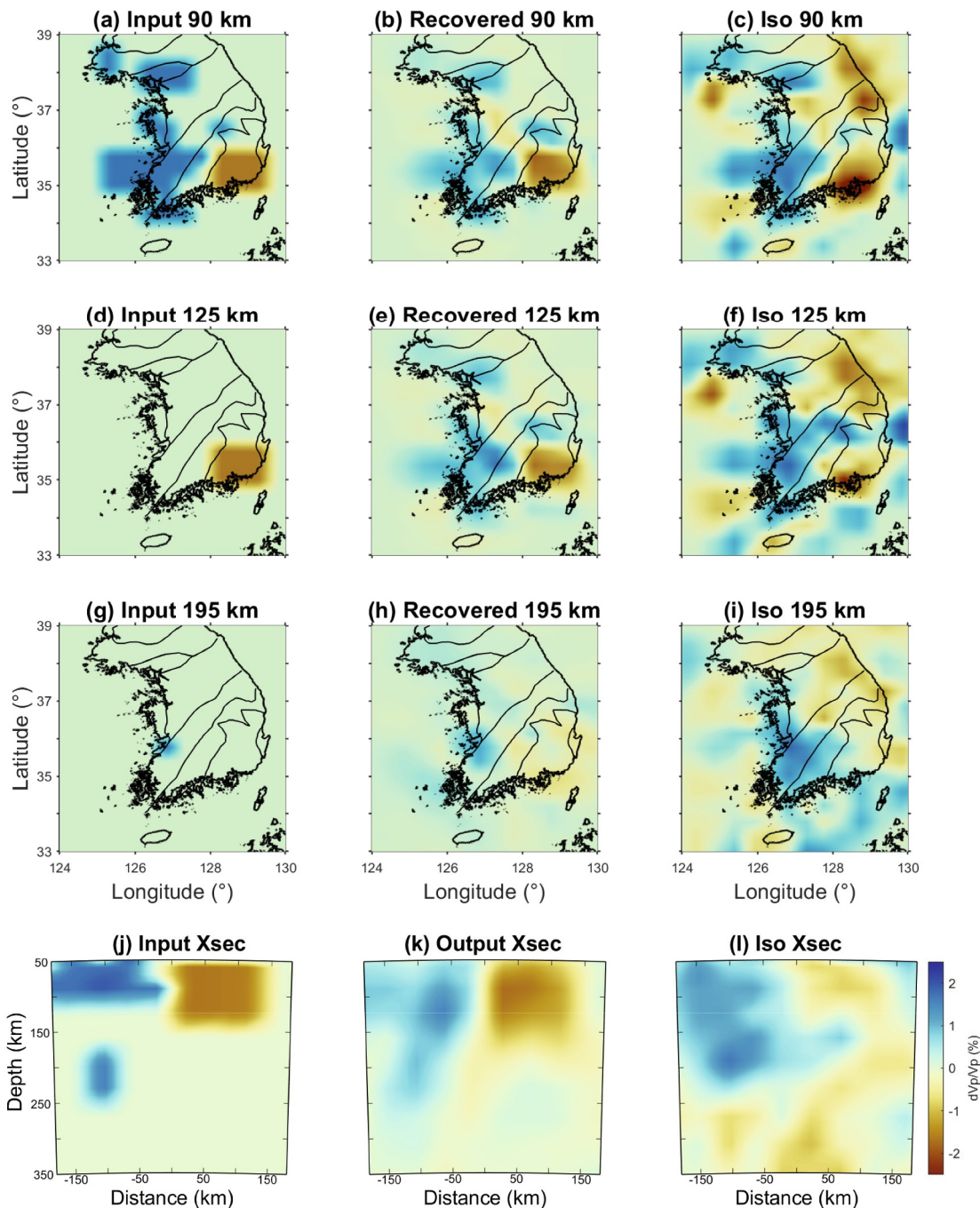


Figure 2. Comparison of synthetic test results (a, b, d, e, g, h, j, k) and isotropic inversion results (c, f, i, l) at depths of 90, 125, and 195 km. Panels (j–l) show a cross section along 35.8°N (126–130°E), marked by the red solid line in (a). The high-velocity delaminated lithosphere ($50 \times 50 \times 75 \text{ km}^3$, shown in (g, j) may be imaged as a deep lithospheric root (h, k).

To assess whether the HVAs beneath the western KP could reflect such ongoing modification, we conducted a suite of synthetic tests designed to examine the resolvability of vertically continuous lithospheric structures (Figure 2, Figures S4 and S17 in Supporting Information S1). A lithospheric lid was imposed between 50 and 100 km depth based on HVA geometry from isotropic tomography. While the LAB in the eastern NCC is estimated to lie at ~ 60 – 70 km depth, we adopted a 50-km-thick lid extending to 100 km, which is compatible with our vertical resolution of ~ 50 km (Figure S2 in Supporting Information S1).

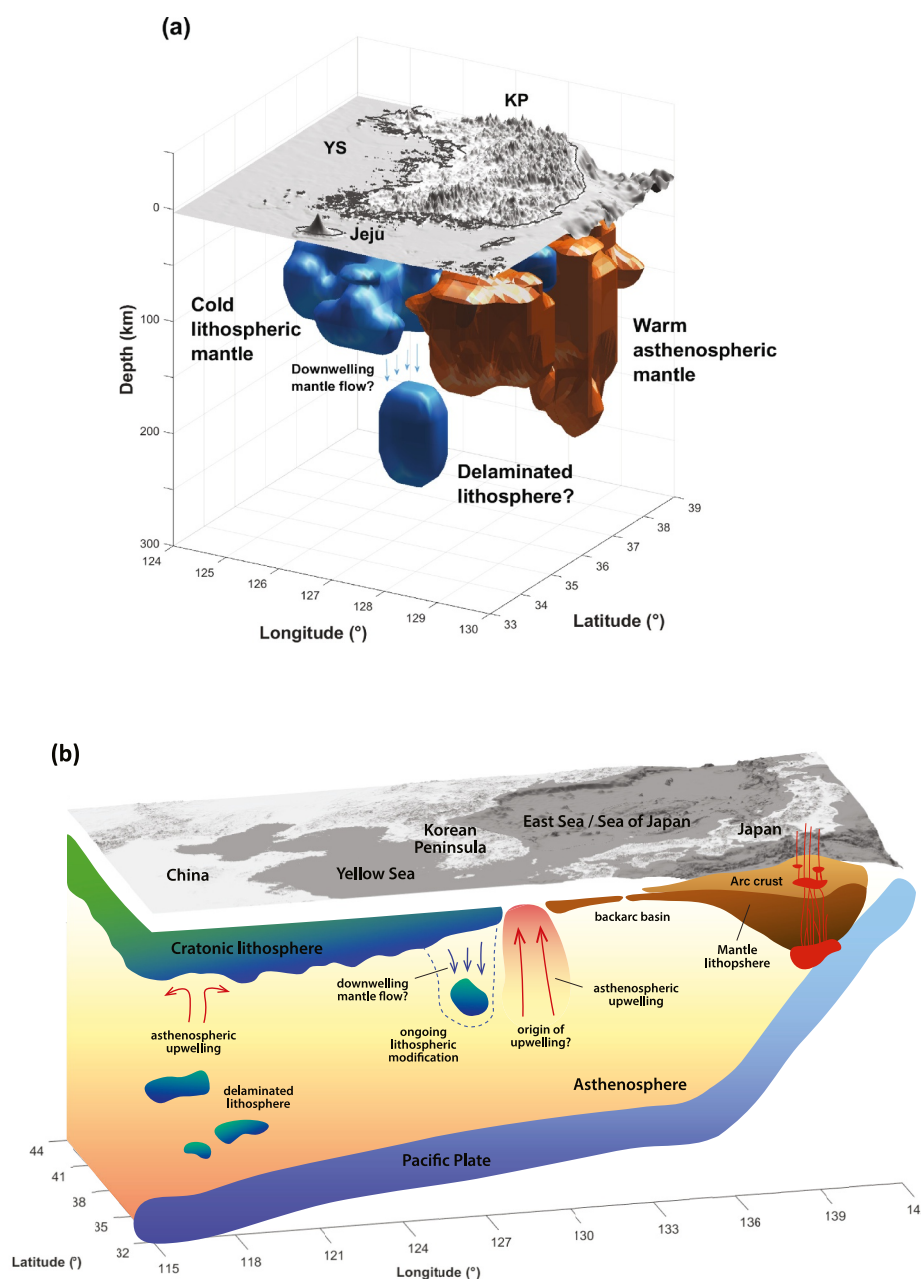


Figure 3. (a) Schematic diagram illustrating possible mantle dynamics beneath the Korean Peninsula (KP). Orange anomalies represent low velocity anomalies that may originate from warm mantle. Meanwhile, blue anomalies in the western KP represent true high velocity anomalies with possible downwelling mantle flow (light blue arrows) and delaminated lithosphere. (b) Conceptual model of mantle processes across Northeast Asia. The illustration highlights lithospheric modification beneath China, situated above the stagnant Pacific Plate subducted from the Japan Trench. The KP lies at the margin of cratonic lithosphere, adjacent to the back-arc opening of the East Sea (Sea of Japan), where mantle structure suggests a transitional stage of lithospheric modification. Hypotheses proposed to explain intraplate volcanism in Northeast Asia (e.g., big mantle wedge or slab tearing) are not included here.

The synthetic tests indicate that both lithospheric hanging and delamination beneath the lithospheric lid can be imaged as a single, contiguous HVA extending to ~ 250 km depth. Perhaps, delamination is more consistent with the modest improvements in the tomography with prescribed anisotropy, which indicates vertically oriented SFD aligned with downwelling flow. Incorporating this anisotropy model yields localized HVA reductions of $\sim 0.5\%$ dV_P/V_P at depths of 125 and 160 km, particularly beneath the southwestern KP (Figure S14 in Supporting

Information S1). Such depth-dependent reductions are consistent vertical mantle flow and alignment of anisotropic fabric. To better discriminate between delamination and a stretched lithosphere, incorporating additional constraints is essential. Including local earthquake data could improve ray geometry and vertical resolution; however, mantle seismicity beneath the Korean Peninsula is rare. Joint or hybrid inversion with other geophysical observations, such as receiver functions and surface-wave dispersion, may provide stronger constraints. Nevertheless, such approaches remain exploratory in this context and would benefit from further validation through geodynamic modeling.

4.3.2. Surface Expression of Lithospheric Modification

If lithospheric delamination is indeed ongoing beneath the western KP, it lacks the typical surface expressions such as topographic uplift or elevated terrain. In well-studied examples such as the Colorado Plateau and the Great Basin, the delamination of lithospheric mantle and the consequent isostatic uplift are often invoked to explain the elevated topography observed in those (He & Kapp, 2023; Levander et al., 2011; Sine et al., 2008; West et al., 2009). Instead, the relatively flat topography and low heat flow values in the western KP (Kim & Lee, 2007) (Figure S18 in Supporting Information S1) are more consistent with hanging or partially detached lithosphere scenarios (Figure S4). In such a scenario, the negative buoyancy of the lithospheric root may promote mantle downwelling while maintaining limited coupling to surface deformation, resulting in subdued or diffuse surface responses.

However, numerical and geodynamic studies demonstrate that lithospheric delamination does not necessarily produce immediate or pronounced surface expressions. In regions where hydration weakens the crust or lithosphere, the resulting surface uplift may be muted, leading to low-relief topography. This framework is relevant to the KP, which resides on the overriding Eurasian Plate and has likely experienced long-term hydration associated with Pacific Plate subduction (He, 2015; Tian & Zhao, 2011, 2013; Yang & Faccenda, 2020; Zhu & Xu, 2019). Other models indicate that strong crustal rheology can suppress isostatic rebound; that is, surface responses depend on rheology, hydration state, viscosity contrasts, and the rate of lithospheric sinking (Beall et al., 2017; Li et al., 2016; McMillan & Schoenbohm, 2023; Molnar & Houseman, 2004). Thus, the lack of prominent topographic uplift across the western KP does not uniquely constrain the style or extent of lithospheric modification at depth (Figure 3).

5. Conclusions

High-resolution P-wave tomography incorporating prescribed mantle anisotropy reveals a pronounced east–west contrast in upper-mantle structure beneath the KP, indicating active lithospheric modification along craton margins in Northeast Asia. LVAs beneath the eastern KP are consistent with elevated temperature and/or hydration, whereas persistent HVAs beneath the western KP reflect a cold lithospheric domain undergoing modification at depth. Tomography with prescribed anisotropy may suggest the presence of vertical mantle flow beneath the western KP, potentially associated with incipient lithospheric delamination, whereas the lack of pronounced surface uplift and low heat flow values favors a hanging lithosphere. These observations suggest a transitional stage of lithospheric modification toward incipient delamination, although further constraints are needed to better resolve its extent and evolution.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Teleseismic data recorded by the KS network are available through the National Earthquake Comprehensive Information System (NECIS; <https://necis.kma.go.kr>). The data from the KG network are accessible via the Big Data Open Platform, KIGAM Quake (<https://data.kigam.re.kr/quake/data/event-data>), operated by the Korea Institute of Geoscience and Mineral Resources (KIGAM). The SWS data were obtained from a previous study (Kang & Shin, 2009). MATLAB R2023b was used for data processing, analysis, and figure generation. The inversion and tomography codes are publicly available through the Stingray/TomoLab package (Toomey & Blue

Tech Seismics, Inc, 2012). The dV_p/V_p values of the isotropic and prescribed anisotropy tomography models are archived at Zenodo (Lee & Kim, 2025).

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2022-NR070842). We sincerely thank the Editor, Dr. Daoyuan Sun, and the two anonymous reviewers for their constructive comments and careful evaluation, which improved the clarity and focus of the manuscript and helped refine our interpretations.

References

- Beall, A. P., Moresi, L., & Stern, T. (2017). Dripping or delamination? A range of mechanisms for removing the lower crust or lithosphere. *Geophysical Journal International*, 210(2), 671–692. <https://doi.org/10.1093/gji/ggx202>
- Bezada, M. J., Faccenda, M., & Toomey, D. R. (2016). Representing anisotropic subduction zones with isotropic velocity models: A characterization of the problem and some steps on a possible path forward: Anisotropic subduction, isotropic models. *Geochemistry, Geophysics, Geosystems*, 17(8), 3164–3189. <https://doi.org/10.1002/2016GC006507>
- Bezada, M. J., Humphreys, E. D., Toomey, D. R., Harnafi, M., Dávila, J. M., & Gallart, J. (2013). Evidence for slab rollback in westernmost Mediterranean from improved upper mantle imaging. *Earth and Planetary Science Letters*, 368, 51–60. <https://doi.org/10.1016/j.epsl.2013.02.024>
- Byrnes, J. S., Bezada, M., Long, M. D., & Benoit, M. H. (2019). Thin lithosphere beneath the central Appalachian Mountains: Constraints from seismic attenuation beneath the MAGIC array. *Earth and Planetary Science Letters*, 519, 297–307. <https://doi.org/10.1016/j.epsl.2019.04.045>
- Cammarano, F., Goes, S., Vacher, P., & Giardini, D. (2003). Inferring upper-mantle temperatures from seismic velocities. *Physics of the Earth and Planetary Interiors*, 138(3), 197–222. [https://doi.org/10.1016/S0031-9201\(03\)00156-0](https://doi.org/10.1016/S0031-9201(03)00156-0)
- Celli, N. L., Lebedev, S., Schaeffer, A. J., & Gaina, C. (2020). African cratonic lithosphere carved by mantle plumes. *Nature Communications*, 11(1), 92. <https://doi.org/10.1038/s41467-019-13871-2>
- Chang, S.-J., & Baag, C.-E. (2005). Crustal structure in Southern Korea from joint analysis of teleseismic receiver functions and surface-wave dispersion. *Bulletin of the Seismological Society of America*, 95(4), 1516–1534. <https://doi.org/10.1785/0120040080>
- Chen, L. (2010). Concordant structural variations from the surface to the base of the upper mantle in the North China Craton and its tectonic implications. *Lithos*, 120(1), 96–115. <https://doi.org/10.1016/j.lithos.2009.12.007>
- Chen, Y., Gu, Y. J., Currie, C. A., Johnston, S. T., Hung, S.-H., Schaeffer, A. J., & Audet, P. (2019). Seismic evidence for a mantle suture and implications for the origin of the Canadian Cordillera. *Nature Communications*, 10(1), 2249. <https://doi.org/10.1038/s41467-019-09804-8>
- Chough, S. K., Kwon, S.-T., Ree, J.-H., & Choi, D. K. (2000). Tectonic and sedimentary evolution of the Korean peninsula: A review and new view. *Earth-Science Reviews*, 52(1), 175–235. [https://doi.org/10.1016/S0012-8252\(00\)00029-5](https://doi.org/10.1016/S0012-8252(00)00029-5)
- Dong, W., Xu, T., Ai, Y., Fan, E., Li, L., & Hou, J. (2022). The boundary between the North China Craton and the Central Asian Orogenic Belt in NE China: Seismic evidence from receiver function imaging. *Journal of Asian Earth Sciences*, 237, 105360. <https://doi.org/10.1016/j.jseaes.2022.105360>
- Feng, J. K., Yao, H. J., Chen, L., & Li, C. L. (2022). Ongoing lithospheric alteration of the North China Craton revealed by surface-wave tomography and geodetic observations. *Geophysical Research Letters*, 49(14), e2022GL099403. <https://doi.org/10.1029/2022GL099403>
- He, C., Dong, S., Santosh, M., Li, Q., & Chen, X. (2015). Destruction of the North China Craton: A perspective based on receiver function analysis. *Geological Journal*, 50(1), 93–103. <https://doi.org/10.1002/gj.2530>
- He, J. J. Y., & Kapp, P. (2023). Basin record of a Miocene lithosphere drip beneath the Colorado Plateau. *Nature Communications*, 14(1), 4433. <https://doi.org/10.1038/s41467-023-40147-7>
- He, L. (2015). Thermal regime of the North China Craton: Implications for craton destruction. *Earth-Science Reviews*, 140, 14–26. <https://doi.org/10.1016/j.earscirev.2014.10.011>
- Hirschmann, M. M. (2000). Mantle solidus: Experimental constraints and the effects of peridotite composition. *Geochemistry, Geophysics, Geosystems*, 1(10). <https://doi.org/10.1029/2000GC000070>
- Ismaïl, W. B., & Mainprice, D. (1998). An olivine fabric database: An overview of upper mantle fabrics and seismic anisotropy. *Tectonophysics*, 296(1–2), 145–157. [https://doi.org/10.1016/S0040-1951\(98\)00141-3](https://doi.org/10.1016/S0040-1951(98)00141-3)
- Kang, T.-S., & Shin, J. S. (2009). Shear-wave splitting beneath southern Korea and its tectonic implication. *Tectonophysics*, 471(3), 232–239. <https://doi.org/10.1016/j.tecto.2009.02.021>
- Karato, S. (1993). Importance of anelasticity in the interpretation of seismic tomography. *Geophysical Research Letters*, 20(15), 1623–1626. <https://doi.org/10.1029/93GL01767>
- Kennett, B., Engdahl, E., & Buland, R. (1995). Constraints on seismic velocities in the Earth from travel-times. *Geophysical Journal International*, 122(1), 108–124. <https://doi.org/10.1111/j.1365-246X.1995.tb03540.x>
- Kim, H., Lee, H., Song, J.-H., Lee, W., Hong, J., Kagoshima, T., et al. (2024). Latent magmatism beneath the Korean Peninsula caused by asthenosphere upwelling. *Communications Earth & Environment*, 5(1), 1–7. <https://doi.org/10.1038/s43247-024-01592-4>
- Kim, H. C., & Lee, Y. (2007). Heat flow in the Republic of Korea. *Journal of Geophysical Research*, 112(B5). <https://doi.org/10.1029/2006JB004266>
- Kim, J.-C. (2002). 1:250,000-scale digital geological map of Korea, Korea Institute of Geoscience and Mineral resources (KIGAM) [Dataset]. <https://doi.org/10.22747/data.20231227.5375>
- Kusky, T. M., Windley, B. F., Wang, L., Wang, Z., Li, X., & Zhu, P. (2014). Flat slab subduction, trench suction, and craton destruction: Comparison of the North China, Wyoming, and Brazilian cratons. *Tectonophysics*, 630, 208–221. <https://doi.org/10.1016/j.tecto.2014.05.028>
- Lee, C.-T. A., Luffi, P., & Chin, E. J. (2011). Building and destroying Continental mantle. *Annual Review of Earth and Planetary Sciences*, 39(39), 59–90. <https://doi.org/10.1146/annurev-earth-040610-133505>
- Lee, H., Bezada, M. J., & Faccenda, M. (2021). Can sub-slab low-velocity anomalies be an artifact caused by anisotropy? A case study from the Alboran slab area in the western Mediterranean. *Tectonophysics*, 819, 229080. <https://doi.org/10.1016/j.tecto.2021.229080>
- Lee, H., & Kim, Y. (2025). Teleseismic P-wave velocity model for the Korean Peninsula (KP) [Dataset]. *Zenodo*. <https://doi.org/10.5281/zenodo.15844674>
- Lee, H., Kim, Y., Bezada, M. J., & Clayton, R. W. (2023). The role of subslab low-velocity anomalies beneath the Nazca Ridge and Iquique Ridge on the Nazca Plate and their possible contribution to the subduction angle. *Geophysical Research Letters*, 50(23), e2023GL106096. <https://doi.org/10.1029/2023GL106096>
- Lee, W., Lee, H., Kim, H., Song, J.-H., Hong, J., Park, J., et al. (2025). Water and gas geochemistry of springs in Ulleungdo volcano, South Korea: Implications for degassing of upper mantle-derived volatiles in Northeast Asia. *Journal of Hydrology*, 647, 132286. <https://doi.org/10.1016/j.jhydrol.2024.132286>
- Lei, J. (2012). Upper-mantle tomography and dynamics beneath the North China Craton. *Journal of Geophysical Research*, 117(B6). <https://doi.org/10.1029/2012JB009212>

- Levander, A., Schmandt, B., Miller, M. S., Liu, K., Karlstrom, K. E., Crow, R. S., et al. (2011). Continuing Colorado plateau uplift by delamination-style convective lithospheric downwelling. *Nature*, 472(7344), 461–465. <https://doi.org/10.1038/nature10001>
- Li, X., Zhu, P., Kusky, T. M., Gu, Y., Peng, S., Yuan, Y., & Fu, J. (2015). Has the Yangtze craton lost its root? A comparison between the North China and Yangtze cratons. *Tectonophysics*, 655, 1–14. <https://doi.org/10.1016/j.tecto.2015.04.008>
- Li, Z., Liu, M., & Gerya, T. (2016). Lithosphere delamination in continental collisional orogens: A systematic numerical study. *Journal of Geophysical Research: Solid Earth*, 121(7), 5186–5211. <https://doi.org/10.1002/2016JB013106>
- McMillan, M., & Schoenbohm, L. M. (2023). Diverse styles of lithospheric dripping: Synthesizing gravitational instability models, Continental tectonics, and geologic observations. *Geochemistry, Geophysics, Geosystems*, 24(2), e2022GC010488. <https://doi.org/10.1029/2022GC010488>
- Molnar, P., & Houseman, G. A. (2004). The effects of buoyant crust on the gravitational instability of thickened mantle lithosphere at zones of intracontinental convergence: Crust and gravitational instability of thickened mantle lithosphere. *Geophysical Journal International*, 158(3), 1134–1150. <https://doi.org/10.1111/j.1365-246X.2004.02312.x>
- Park, S., & Hong, T.-K. (2024). Continent-Side uplifted mantle and geological imprints along a Paleo rift in the Western East Sea (Sea of Japan). *Journal of Geophysical Research: Solid Earth*, 129(9), e2024JB029049. <https://doi.org/10.1029/2024JB029049>
- Polat, A., Li, J., Fryer, B., Kusky, T., Gagnon, J., & Zhang, S. (2006). Geochemical characteristics of the Neoproterozoic (2800–2700 Ma) Taishan greenstone belt, North China Craton: Evidence for plume–craton interaction. *Chemical Geology*, 230(1), 60–87. <https://doi.org/10.1016/j.chemgeo.2005.11.012>
- Sine, C. R., Wilson, D., Gao, W., Grand, S. P., Aster, R., Ni, J., & Baldrige, W. S. (2008). Mantle structure beneath the western edge of the Colorado Plateau. *Geophysical Research Letters*, 35(10). <https://doi.org/10.1029/2008GL033391>
- Song, J.-H., Kim, S., & Rhie, J. (2020). Heterogeneous modification and reactivation of a craton margin beneath the Korean Peninsula from teleseismic travel time tomography. *Gondwana Research*, 81, 475–489. <https://doi.org/10.1016/j.gr.2019.11.016>
- Tatsumi, Y., Otofujii, Y.-I., Matsuda, T., & Nohda, S. (1989). Opening of the Sea of Japan back-arc basin by asthenospheric injection. *Tectonophysics*, 166(4), 317–329. [https://doi.org/10.1016/0040-1951\(89\)90283-7](https://doi.org/10.1016/0040-1951(89)90283-7)
- Tian, Y., & Zhao, D. (2011). Destruction mechanism of the North China Craton: Insight from P and S wave mantle tomography. *Journal of Asian Earth Sciences*, 42(6), 1132–1145. <https://doi.org/10.1016/j.jseae.2011.06.010>
- Tian, Y., & Zhao, D. (2013). Reactivation and mantle dynamics of North China Craton: Insight from P-wave anisotropy tomography. *Geophysical Journal International*, 195(3), 1796–1810. <https://doi.org/10.1093/gji/ggt333>
- Toomey, D. R., & Blue Tech Seismics, Inc. (2012). Stingray/TomoLab. Retrieved from <https://pages.uoregon.edu/drt/Stingray/pages/89.html>
- VanDecar, J. C., & Crosson, R. S. (1990). Determination of teleseismic relative phase arrival times using multi-channel cross-correlation and least squares. *Bulletin of the Seismological Society of America*, 80(1), 150–169.
- VanderBeek, B. P., & Faccenda, M. (2021). Imaging upper mantle anisotropy with teleseismic P-wave delays: Insights from tomographic reconstructions of subduction simulations. *Geophysical Journal International*, 225(3), 2097–2119. <https://doi.org/10.1093/gji/ggab081>
- Van Hone, A., Sato, H., & Ishiyama, T. (2017). Evolution of the Sea of Japan back-arc and some unsolved issues. *Tectonophysics*, 710–711, 6–20. <https://doi.org/10.1016/j.tecto.2016.08.020>
- Wang, C., Liu, Z.-H., Liu, G., Dong, X.-J., Liu, S.-C., & Zhao, Q.-Y. (2022). Geochronology and geochemistry of Late Triassic intrusions in the Liaodong Peninsula, eastern North China Craton: Implications for post-collisional lithospheric thinning. *International Geology Review*, 64(8), 1033–1050. <https://doi.org/10.1080/00206814.2021.1904295>
- West, J. D., Fouch, M. J., Roth, J. B., & Elkins-Tanton, L. T. (2009). Vertical mantle flow associated with a lithospheric drip beneath the Great Basin. *Nature Geoscience*, 2(6), 439–444. <https://doi.org/10.1038/ngeo526>
- Wu, F.-Y., Yang, J.-H., Xu, Y.-G., Wilde, S. A., & Walker, R. J. (2019). Destruction of the North China Craton in the Mesozoic. *Annual Review of Earth and Planetary Sciences*, 47(47), 173–195. <https://doi.org/10.1146/annurev-earth-053018-060342>
- Xia, B., Thybo, H., & Artemieva, I. M. (2020). Lithosphere mantle density of the North China Craton. *Journal of Geophysical Research: Solid Earth*, 125(9), e2020JB020296. <https://doi.org/10.1029/2020JB020296>
- Xiao, Z., Sun, X., Yang, J., & Gao, Y. (2023). Dense-array adjoint tomography reveals lithospheric delamination and asthenosphere upwelling beneath the western Yangtze Craton. *Frontiers in Earth Science*, 11, 1123633. <https://doi.org/10.3389/feart.2023.1123633>
- Xu, Y., Zeyen, H., Hao, T., Santosh, M., Li, Z., Huang, S., & Xing, J. (2016). Lithospheric structure of the North China Craton: Integrated gravity, geoid and topography data. *Gondwana Research*, 34, 315–323. <https://doi.org/10.1016/j.gr.2015.03.010>
- Yang, J., & Faccenda, M. (2020). Intraplate volcanism originating from upwelling hydrous mantle transition zone. *Nature*, 579(7797), 88–91. <https://doi.org/10.1038/s41586-020-2045-y>
- Zandt, G., Gilbert, H., Owens, T. J., Ducea, M., Saleeby, J., & Jones, C. H. (2004). Active foundering of a continental arc root beneath the southern Sierra Nevada in California. *Nature*, 431(7004), 41–46. <https://doi.org/10.1038/nature02847>
- Zhang, H., Lü, Q.-T., Wang, X.-L., Han, S., Liu, L., Gao, L., et al. (2023). Seismically imaged lithospheric delamination and its controls on the Mesozoic Magmatic Province in South China. *Nature Communications*, 14(1), 2718. <https://doi.org/10.1038/s41467-023-37855-5>
- Zhang, P., Yao, H., Chen, L., Fang, L., Wu, Y., & Feng, J. (2019). Moho depth variations from receiver function imaging in the Northeastern North China Craton and its tectonic implications. *Journal of Geophysical Research: Solid Earth*, 124(2), 1852–1870. <https://doi.org/10.1029/2018JB016122>
- Zhang, S.-H., Zhao, Y., Davis, G. A., Ye, H., & Wu, F. (2014). Temporal and spatial variations of Mesozoic magmatism and deformation in the North China Craton: Implications for lithospheric thinning and decratonization. *Earth-Science Reviews*, 131, 49–87. <https://doi.org/10.1016/j.earscirev.2013.12.004>
- Zhou, Y., & Zhai, M. (2022). Mantle plume-triggered rifting closely following Neoproterozoic cratonization revealed by 2.50–2.20 Ga magmatism across North China Craton. *Earth-Science Reviews*, 230, 104060. <https://doi.org/10.1016/j.earscirev.2022.104060>
- Zhu, R., & Xu, Y. (2019). The subduction of the west Pacific plate and the destruction of the North China Craton. *Science China Earth Sciences*, 62(9), 1340–1350. <https://doi.org/10.1007/s11430-018-9356-y>
- Zhu, R.-X., Yang, J.-H., & Wu, F.-Y. (2012). Timing of destruction of the North China Craton. *Lithos*, 149, 51–60. <https://doi.org/10.1016/j.lithos.2012.05.013>